

GLOBAL AI ADOPTION IN EDUCATION

Tableau Dashboard, Data Story and Flask Web Integration

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Abstract. This report documents a complete data-visualization project on AI adoption in education across ten countries from 2015 to 2026. A CSV dataset of 1,360 records was prepared and analysed in Tableau, producing six visualizations, one interactive dashboard and a six-scene data story, which were then embedded in a Flask web application. Sections 1 to 5 follow the project phase templates (planning, data preparation, visualization, dashboard, story) and Section 6 covers web integration, key insights and conclusion.

1. Project Initialization and Planning Phase

Define Problem Statements | Proposed Solution | Initial Project Planning

1.1 Define Problem Statements

The customer problem statements below describe the viewpoint of the people who would use the dashboard.

PS	I am (customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An education decision maker	understand how fast AI adoption is growing in schools	I only find scattered survey tables	the data is not presented visually or over time	unsure about planning and budgets
PS-2	A teacher	compare my AI usage with students and see which tool is popular	I cannot see a simple comparison	reports mix many indicators in raw numbers	left behind and confused about which tool to adopt
PS-3	A student or researcher	compare countries and rural vs urban usage	comparisons are hard across countries and years	there is no single interactive view	frustrated and unable to draw conclusions

1.2 Proposed Solution

Objective	Present AI adoption in education (2015-2026) through clear Tableau visualizations, one interactive dashboard and a data story, made accessible from a Flask web page.
Scope	Ten countries, 12 years, student/teacher/urban/rural usage, AI tool popularity and supporting indicators. Uses a sample/synthetic CSV dataset.
Problem description	AI-in-education information is scattered across surveys and tables, which makes trends and comparisons hard to see.
Impact of solving	Faster, evidence-based understanding of growth, gaps (rural vs urban, students vs teachers) and tool preferences.
Approach	CSV dataset → Tableau import → data preparation → 6 visualizations → dashboard → 6-scene story → Tableau Public → Flask embedding.
Key features	Interactive filters (Country, Year, Top AI Tool); tooltips; story navigation; web page with Dashboard and Story buttons.

Resource requirements

Resource type	Description	Specification / allocation
Hardware	Computing / memory / storage	Standard laptop or PC, 8 GB RAM, no GPU needed; minimal disk space for CSV and workbook
Software	BI tool and hosting	Tableau Desktop / Tableau Public (publishing)
Software	Web framework and languages	Python Flask, HTML, CSS, JavaScript
Software	Environment	Any code editor (e.g. VS Code), web browser
Data	Source, size, format	Global AI in Education - CSV, 1,360 rows x 16 columns

1.3 Initial Project Planning (Product Backlog and Sprint Schedule)

Sprint	Epic	ID	User story / task	Pts	Priority	Start	End
Sprint-1	Data preparation	USN-1	As an analyst, I can load the CSV into Tableau and verify data types	2	High	[date]	[date]
Sprint-1	Data preparation	USN-2	As an analyst, I can check missing values and create calculated fields	2	High	[date]	[date]
Sprint-2	Visualization	USN-3	As a user, I can view six charts (trend, country, student usage, student vs teacher, rural, tools)	5	High	[date]	[date]
Sprint-2	Dashboard	USN-4	As a user, I can see all key charts on one interactive dashboard with filters	3	High	[date]	[date]
Sprint-3	Story	USN-5	As a user, I can follow a six-scene story from overview to findings	3	Medium	[date]	[date]
Sprint-3	Web integration	USN-6	As a user, I can open the Dashboard and Story from a Flask web page	3	High	[date]	[date]
Sprint-4	Documentation	USN-7	As a reviewer, I can read a complete report of the project	2	Medium	[date]	[date]

Team member for all tasks: Sneha Yadav. Sprint dates are to be filled in by the student.

2. Data Collection and Preprocessing Phase

Raw Data Sources | Data Quality Report | Data Exploration and Preprocessing

2.1 Data Collection Plan and Raw Data Sources

The project uses a single CSV file containing country-wise, year-wise and month-wise records of AI usage in education. The data is sample/synthetic and was created for learning and demonstration.

Source name	Description	Location	Format	Size	Access
Global AI in Education	AI usage by students, teachers, urban and rural areas, school adoption, internet penetration, education index, top AI tool	Local file: Global_AI_in_Education.csv	CSV	1,360 rows x 16 columns	Public / provided for the project

2.2 Data Quality Report

Data source	Data quality issue	Severity	Resolution plan
Global AI in Education	Column government_ai_policy has 441 missing values (32.4% of rows). All other columns are complete.	Moderate	Label blanks as "Not Specified" or exclude the field; it is not used in the six visualizations.
Global AI in Education	Duplicate records: none found (0 duplicates).	Low	No action needed.
Global AI in Education	Data is synthetic; country averages are almost identical (30.6% - 31.6%).	Moderate	State this limitation clearly; do not treat results as real-world statistics.
Global AI in Education	Year, month and percentage values are stored as numbers; Year must behave as a dimension in charts.	Low	Set Year as a discrete dimension in Tableau; keep percentages as measures.

2.3 Data Exploration and Preprocessing

Section	What was done
Data overview	1,360 records, 16 columns; 10 countries, 6 regions, years 2015-2026 (10 countries per month, 136 country-month groups).
Data cleaning	Checked for missing values (only government_ai_policy, 441) and duplicates (none). No rows removed.
Data transformation	Aggregated measures (average student AI usage by year, country and tool); created calculated fields such as Adoption Category and Impact Category; applied Country, Year and Top AI Tool filters.
Data type conversion	Country, Region, Top AI Tool, Government AI Policy, AI in Curriculum treated as dimensions; Year as discrete dimension; usage percentages, index and minutes as measures.
Column splitting / merging	Not required - each field already holds a single value.
Data modeling	Single-table data source; no joins or relationships needed. Measures are aggregated by Tableau (average).
Save processed data	Prepared data kept in the Tableau workbook and published to Tableau Public as "Global ai Adoption in Education Final".

Data dictionary

Column	Type	Meaning
year, month	Dimension / numeric	Year (2015-2026) and month of the record
country, region	Dimension	Country (10) and its region (6)
student_ai_usage_pct / teacher_ai_usage_pct	Measure	Percentage of students / teachers using AI
schools_ai_adoption_pct	Measure	Percentage of schools that adopted AI
avg_daily_ai_usage_min	Measure	Average minutes of AI use per day
top_ai_tool	Dimension	ChatGPT, Google Gemini, Khanmigo or Microsoft Copilot
internet_penetration_pct, education_index	Measure	Internet access (%) and education level (0-1)
government_ai_policy, ai_in_curriculum	Dimension	Policy status (Active/Draft) and AI in curriculum (Yes/No)
urban_ai_usage_pct / rural_ai_usage_pct	Measure	AI usage in urban / rural areas
gender_gap_ai_usage_pct	Measure	Gap in AI usage between genders

Summary of key measures (all records)

Measure	Student	Teacher	Urban	Rural	Schools adoption	Daily minutes	Education index
Average	31.0%	29.9%	33.6%	21.6%	29.4%	32.8	0.78

3. Data Visualization

Business Questions and Visualization Report (minimum 8 required)

Eight business questions guide the visualizations. Questions 1-6 use the Tableau worksheets; questions 4 and 5 are also supported by supplementary year-wise charts (marked "supplementary") built from the same CSV to give a clearer comparison than the summed bars in Tableau.

Business Question 1: How has AI adoption among students changed over the years?

Visualization: Line chart - Year vs Average Student AI Usage Pct.

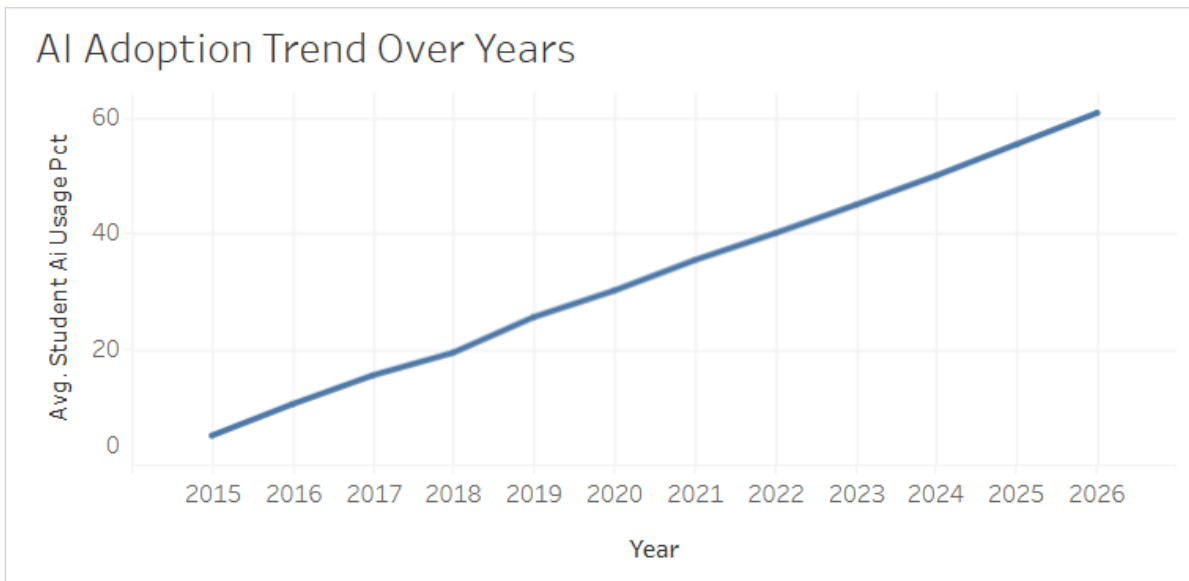


Figure 1: AI Adoption Trend Over Years

Insight: Average student usage rises every year, from about 5% in 2015 to about 61% in 2026 - roughly 5 percentage points per year with no decline.

Business Question 2: Which countries use AI the most?

Visualization: Bar chart - Average Student AI Usage Pct by Country.

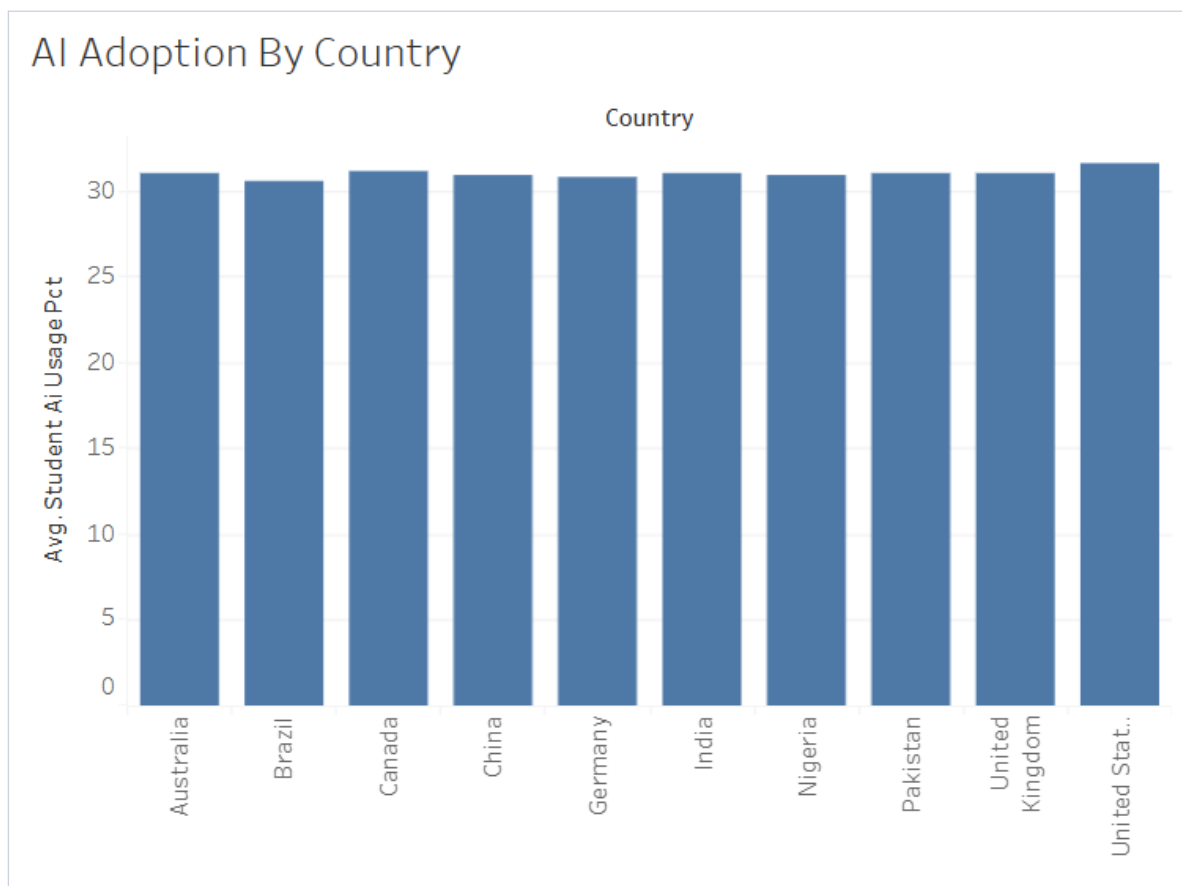


Figure 2: AI Adoption By Country

Insight: All ten countries are very close (about 30.6% to 31.6%). The United States is marginally highest and Brazil marginally lowest.

Business Question 3: How does overall student AI usage relate to the education index?

Visualization: Scatter plot - Average Education Index vs Average Student AI Usage Pct.

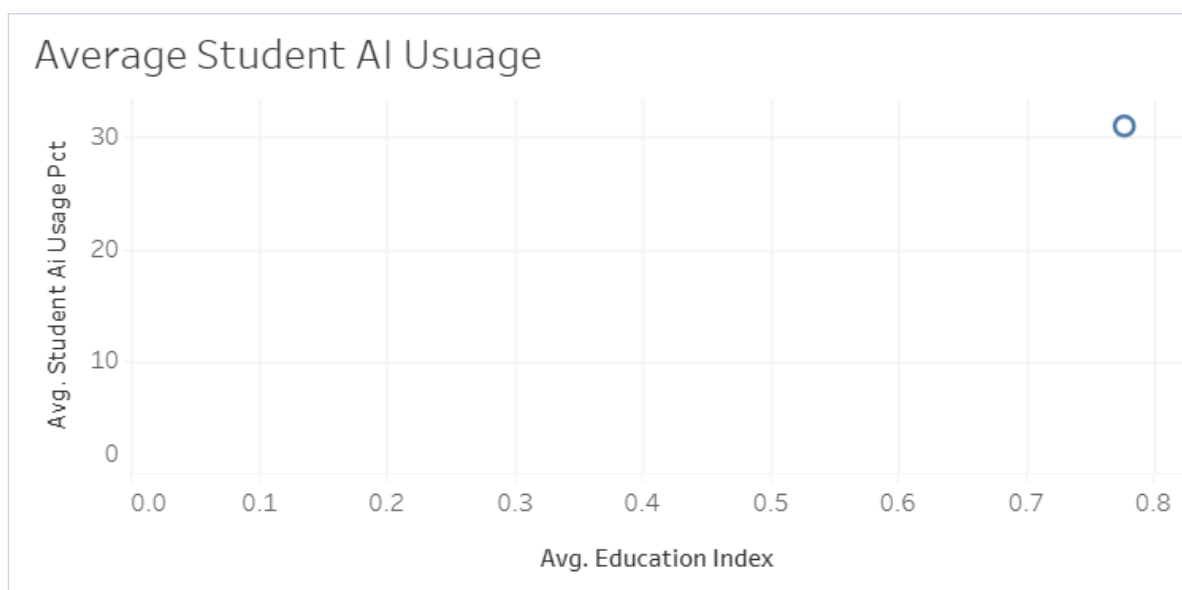


Figure 3: Average Student AI Usage

Insight: Because both values are averaged over the whole dataset, one summary point appears: education index about 0.78 and student usage about 31%.

Business Question 4: Do students use AI more than teachers?

Visualization: Bar chart of measure values (Tableau) plus supplementary line chart by year.

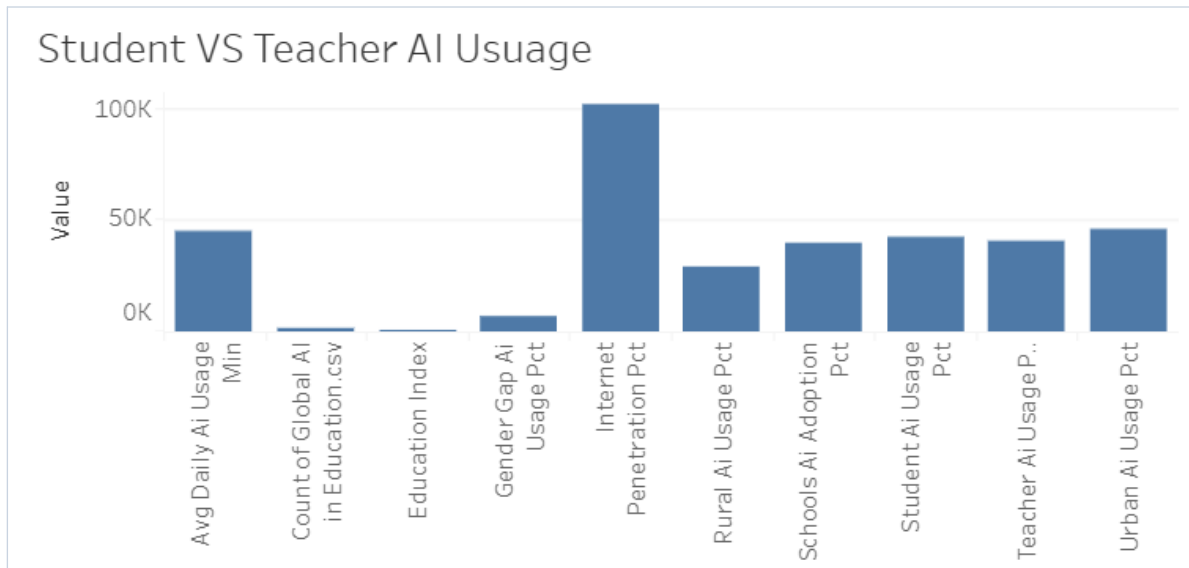


Figure 4: Student vs Teacher AI Usage

Insight: Students (31.0% average) use AI slightly more than teachers (29.9%). Bars in Figure 4 are summed values, so read them as relative comparisons only.

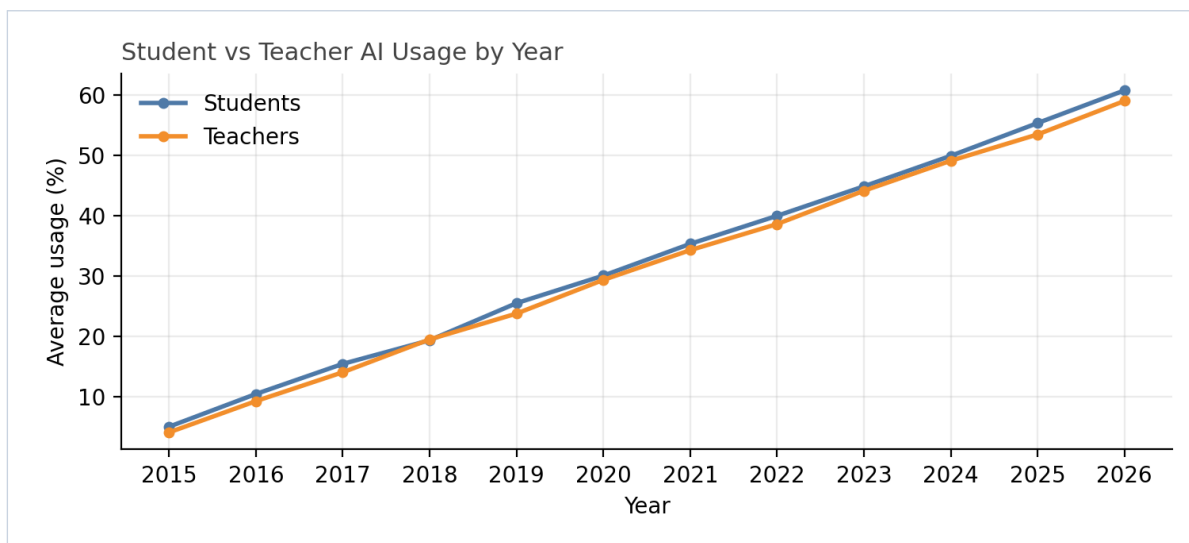


Figure 4a (supplementary): Student vs Teacher AI Usage by Year

Insight: The two lines move together; students stay ahead of teachers in every year except 2018, and the gap is always small (about 1-2 points).

Business Question 5: Is rural AI usage lower than urban usage?

Visualization: Scatter plot - Average Rural AI Usage Pct vs Average Daily AI Usage (minutes), plus supplementary urban vs rural line chart.

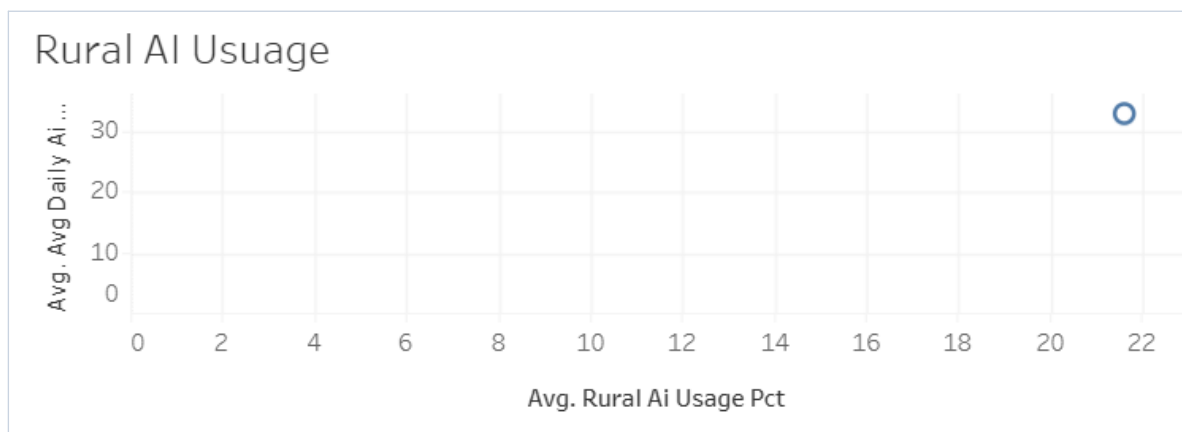


Figure 5: Rural AI Usage

Insight: Rural usage averages about 21.6% against 33.6% for urban areas, with about 33 minutes of AI use per day - a clear rural-urban gap.

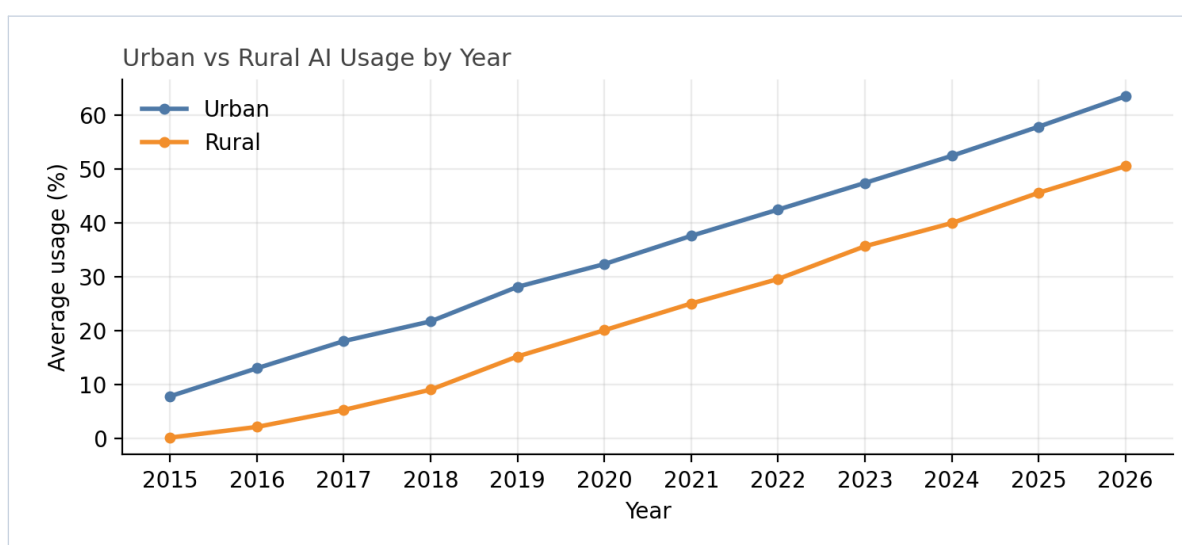


Figure 5a (supplementary): Urban vs Rural AI Usage by Year

Insight: Both rise every year, but rural usage stays well below urban usage - the gap is about 12-13 points in every year from 2017 onward.

Business Question 6: Which AI tool is associated with the highest usage?

Visualization: Horizontal bar chart - Average Student AI Usage Pct by Top AI Tool.

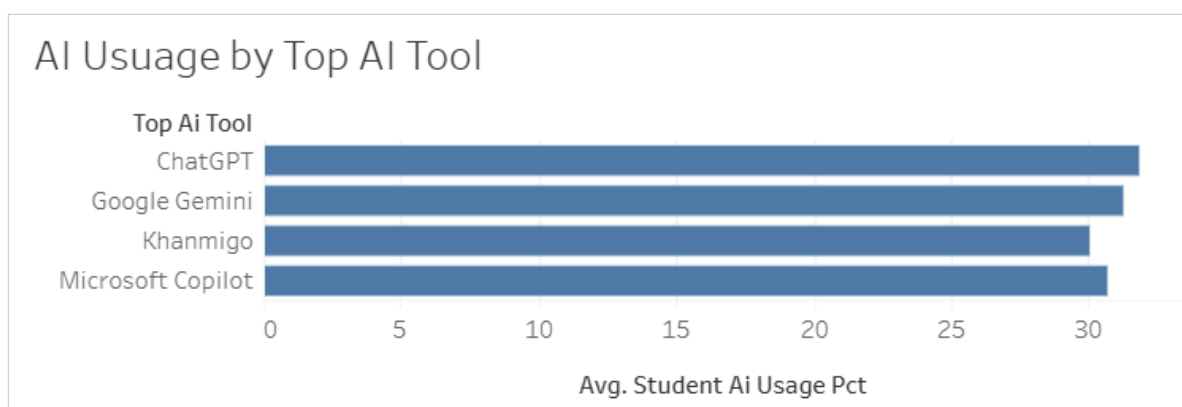


Figure 6: AI Usage by Top AI Tool

Insight: ChatGPT leads (about 31.8%), followed by Google Gemini (31.3%), Microsoft Copilot (30.7%) and Khanmigo (30.0%). The tools are close, so no tool dominates.

Business Questions 7 and 8 (supplementary)

No.	Business question	Visualization	Answer from the data
7	Do countries with higher internet penetration or education index use AI more?	Scatter / bar by country (supplementary analysis)	Internet penetration ranges from 42% (Nigeria) to 96% (Australia), yet student usage stays about 31% everywhere - correlation is near zero in this dataset.
8	Does having AI in the curriculum or an active government policy change usage?	Bar chart by curriculum / policy (supplementary analysis)	Curriculum Yes = 31.3% vs No = 30.6%; policy Active = 30.4% vs Draft = 32.0%. Differences are small (policy has 441 blank values).

4. Dashboard

Dashboard Design | Activity 1: interactive dashboard | Activity 2: publish on Tableau Public

Activity 1: Interactive and visually appealing dashboard

The dashboard combines the six worksheets on one screen so that growth, country comparison, student/teacher usage, rural usage and tool popularity can be read together. Design choices: a clear two-column layout, a single blue colour theme, the right chart type for each question, and interactive filters.

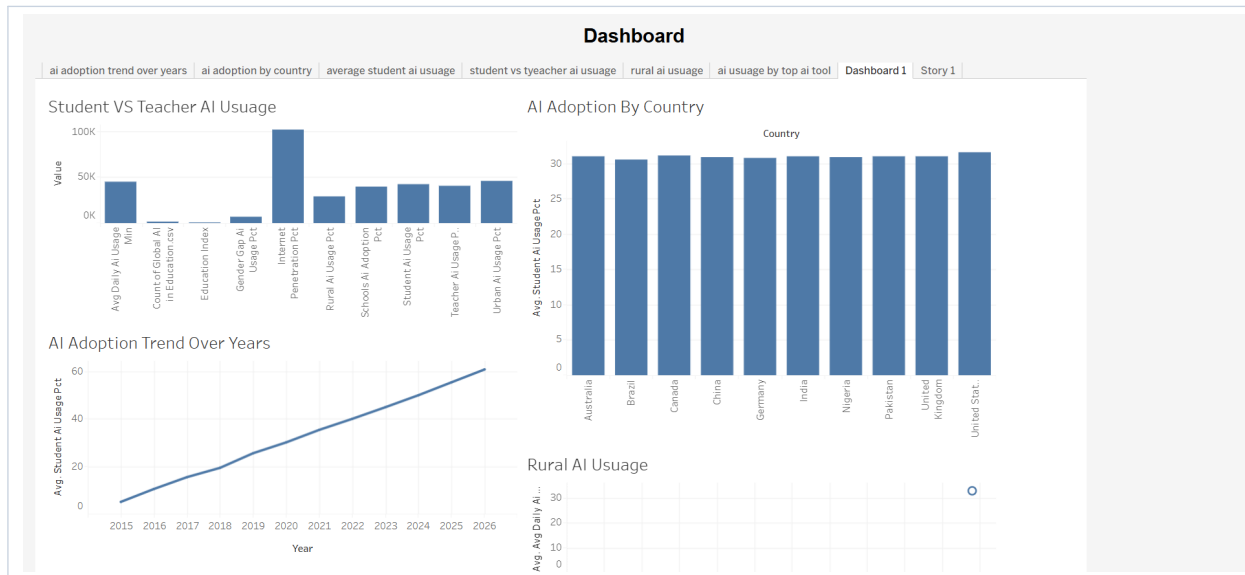


Figure 7: Tableau Dashboard (top section)

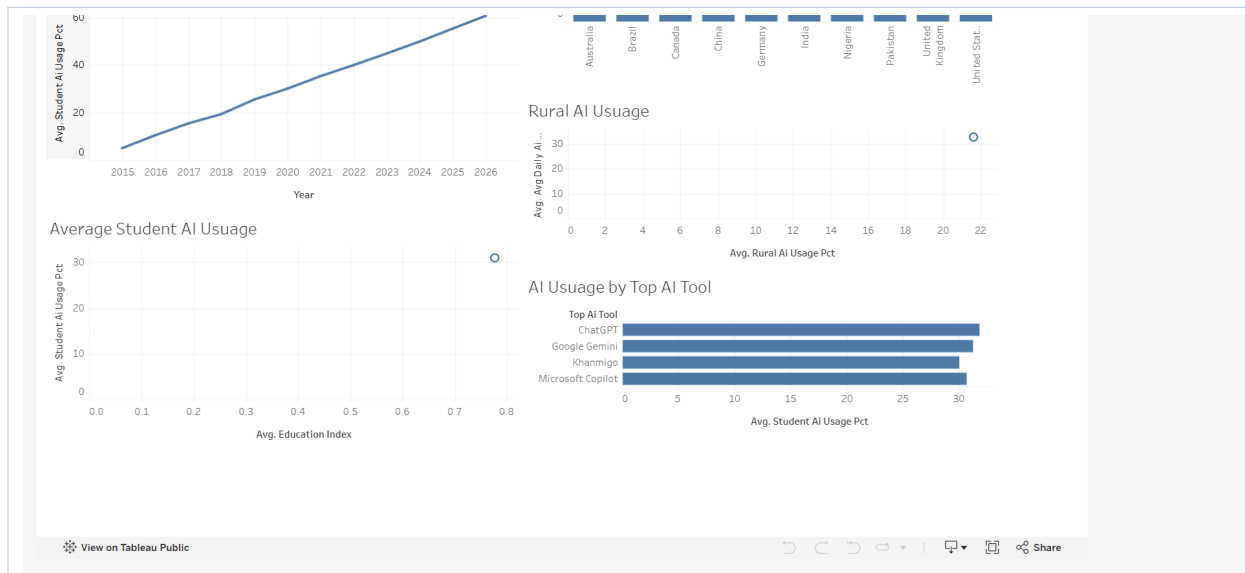


Figure 8: Tableau Dashboard (lower section, published on Tableau Public)

Design features

- **Layout:** related charts grouped on one screen with sheet tabs for navigation.
- **Visualizations:** line for trend, bars for comparisons, scatter for relationships.
- **Colour and theme:** consistent blue palette; no distracting colours.
- **Interactivity:** Country, Year and Top AI Tool filters; tooltips show exact values on hover.

- **Calculated fields:** Adoption Category and Impact Category support the analysis.

Major outcomes from the dashboard

- **Steady growth:** average student AI usage climbs from about 5% (2015) to about 61% (2026).
- **Even spread across countries:** all ten countries average about 31%; the United States is marginally highest (31.6%).
- **Students vs teachers:** student usage (31.0%) is slightly above teacher usage (29.9%).
- **Rural gap:** rural usage averages 21.6% versus 33.6% urban.
- **Tool leader:** ChatGPT ranks first (31.8%), with Gemini, Copilot and Khanmigo close behind.

Activity 2: Dashboard published on Tableau Public

Public link: **[Paste your Tableau Public Dashboard link here]** (workbook: "Global ai Adoption in Education Final" by Sneha Yadav).

5. Story

Story Design | Activity 1: data story | Activity 2: publish on Tableau Public

The Tableau Story arranges the analysis into six scenes, moving from an executive overview to the individual findings.

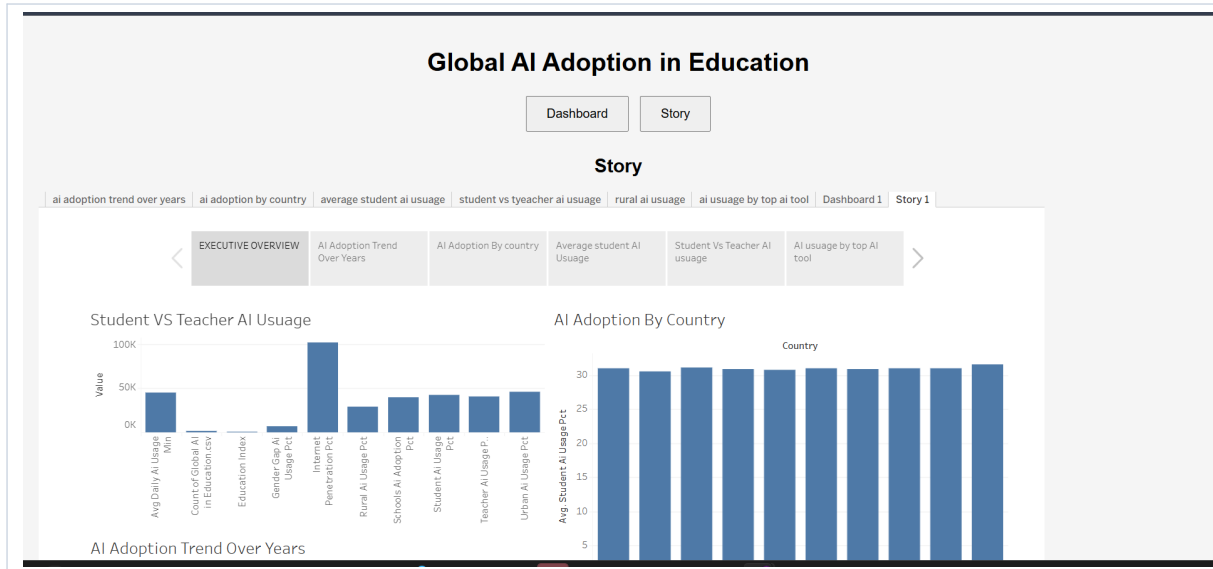


Figure 9: Scene 1 - Executive Overview

Explanation: Gives the complete picture at a glance: student vs teacher comparison, country comparison, adoption trend and other key charts.

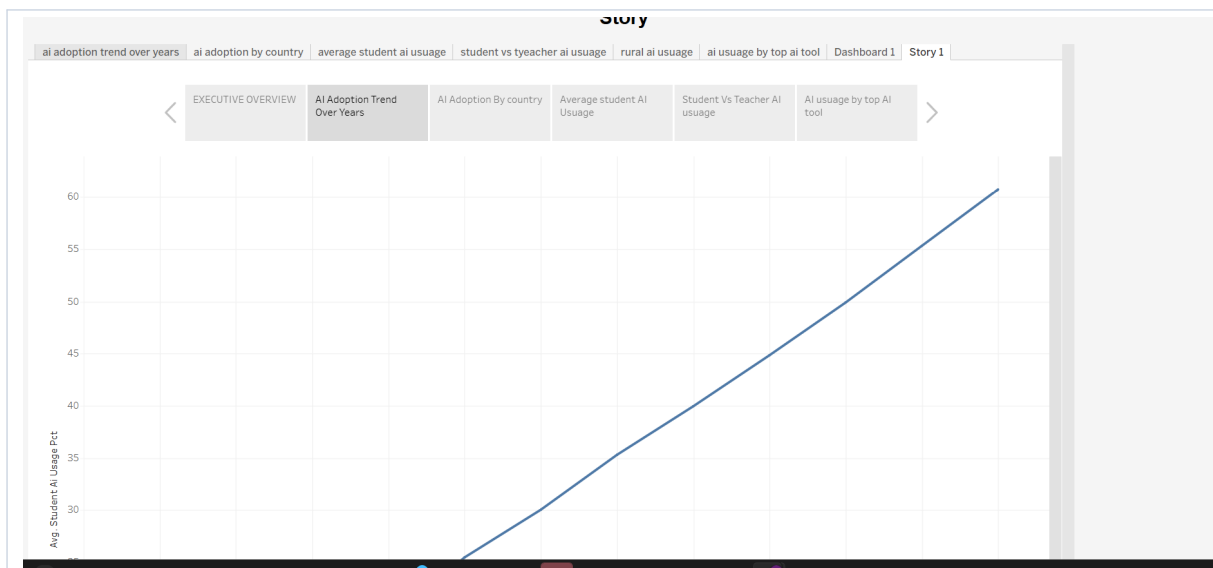


Figure 10: Scene 2 - AI Adoption Trend Over Years

Explanation: Average student AI usage grows steadily from about 5% (2015) to over 60% (2026).

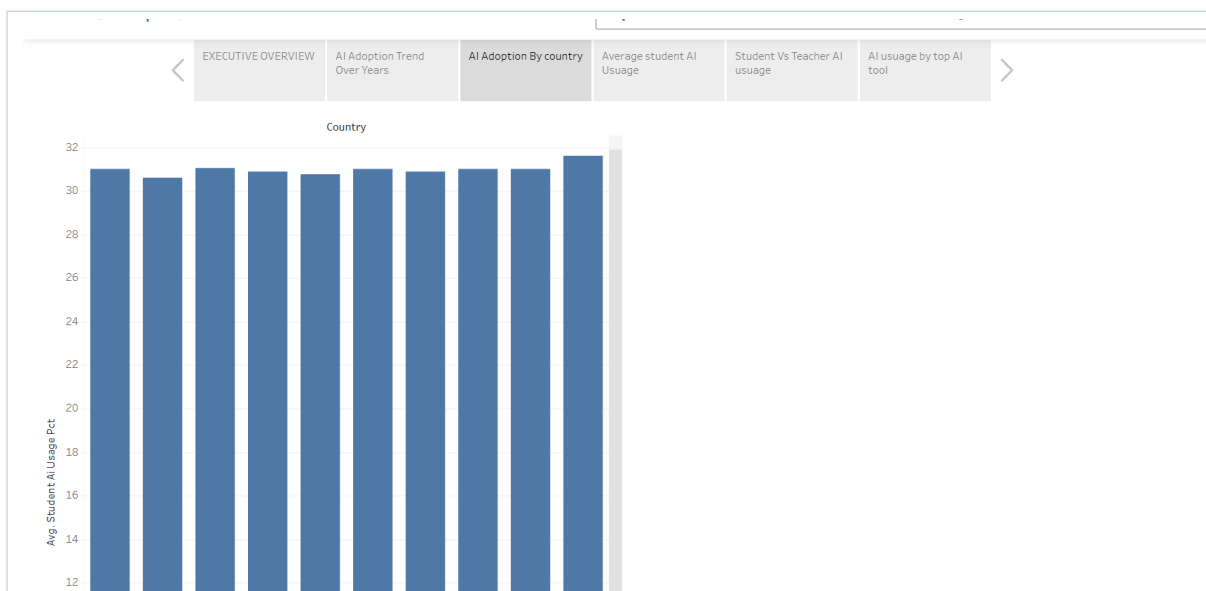


Figure 11: Scene 3 - AI Adoption By Country

Explanation: All ten countries stay close to 31%, with the United States slightly ahead.

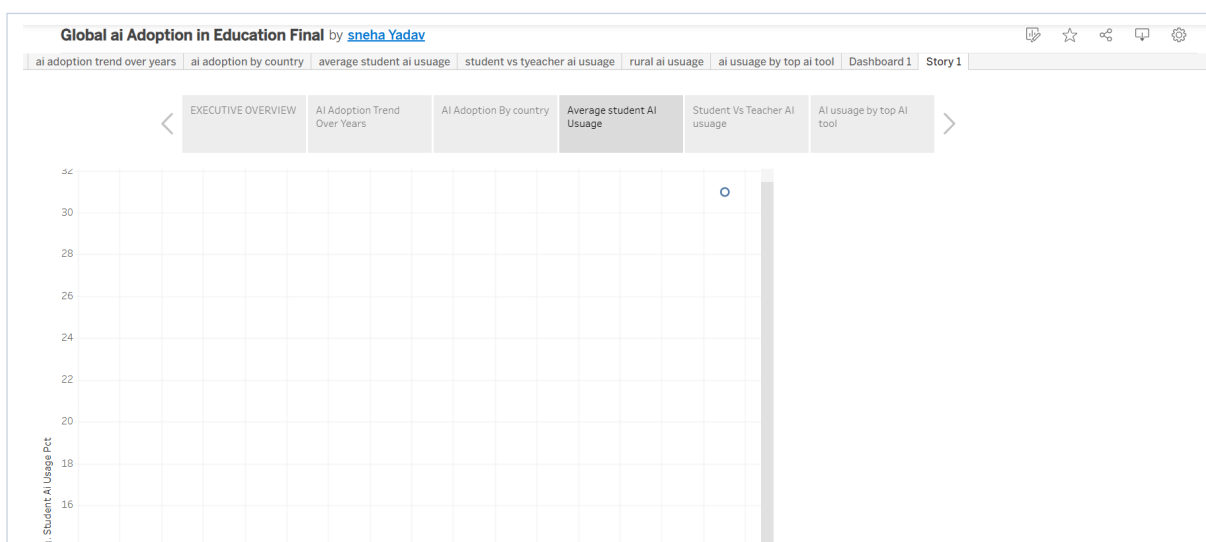


Figure 12: Scene 4 - Average Student AI Usage

Explanation: Overall student AI usage (about 31%) shown against the average education index (about 0.78).

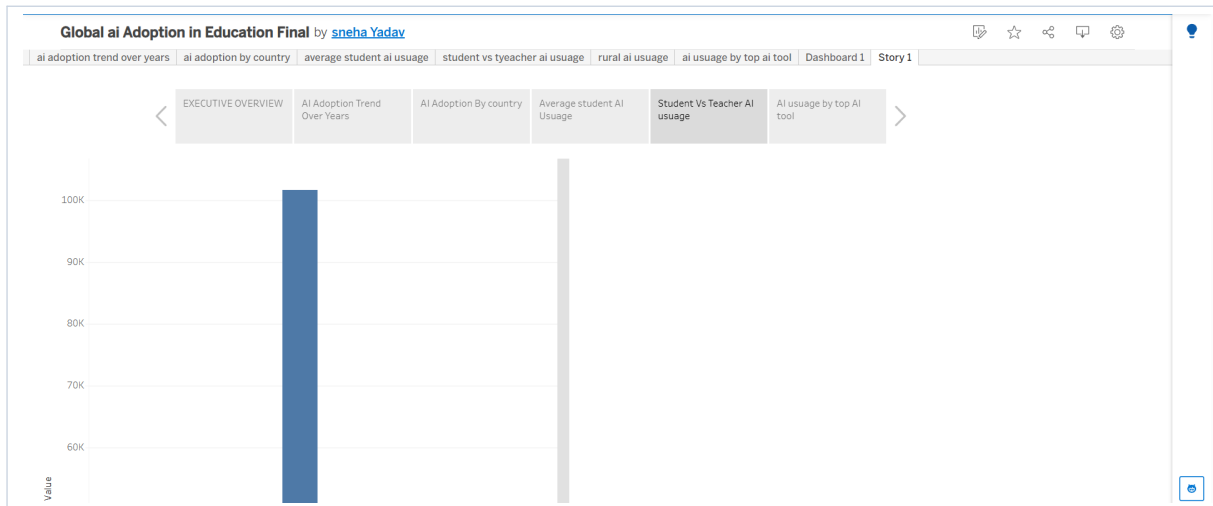


Figure 13: Scene 5 - Student vs Teacher AI Usage

Explanation: Compares student and teacher usage with other indicators; student usage is slightly higher.

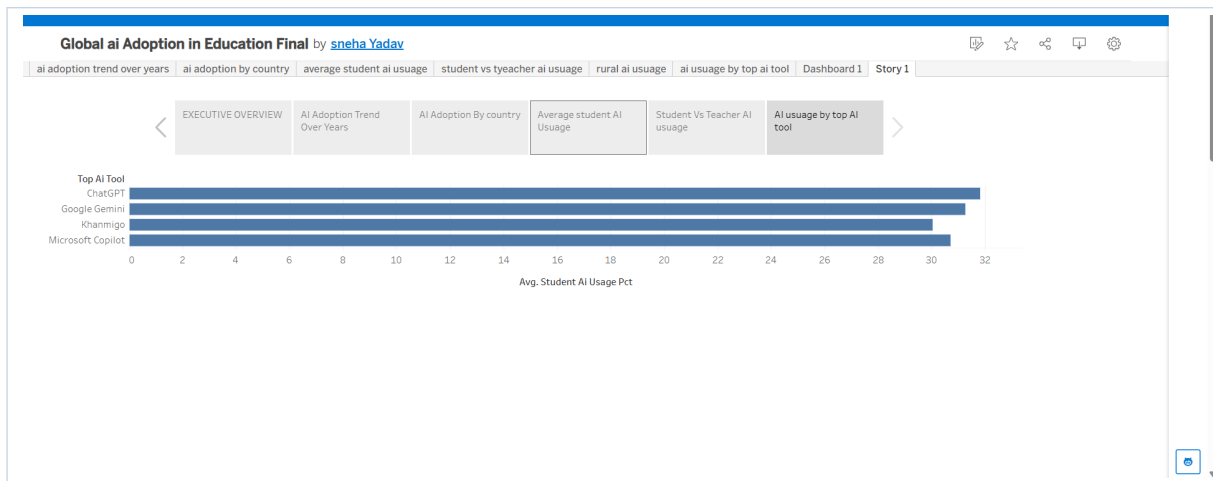


Figure 14: Scene 6 - AI Usage by Top AI Tool

Explanation: Ranks the four tools: ChatGPT leads, followed by Google Gemini, Microsoft Copilot and Khanmigo.

Observations

- **Consistent growth:** about 5 percentage points of student usage are added each year.
- **No country stands out:** averages differ by less than 1 percentage point.
- **Urban vs rural:** rural usage trails urban usage by about 12 points on average.
- **Students vs teachers:** a small but steady student lead.
- **Tools:** ChatGPT is the most used tool, but the field is close.

Activity 2: Story published on Tableau Public

Public link: **[Paste your Tableau Public Story link here]**

6. Flask Web Integration, Results and Conclusion

Web application | Key insights | Limitations | Conclusion | Future scope | References

6.1 Flask Web Integration

A Flask web application was developed to bring the published Tableau Dashboard and Story together. The published views are embedded using HTML, CSS and JavaScript, and the page provides separate Dashboard and Story buttons.

Component	Technology	Purpose
Back end	Python Flask	Serves the web page
Front end	HTML, CSS, JavaScript	Page layout, buttons, switching between views
Visual content	Tableau Public embed	Displays the published Dashboard and Story

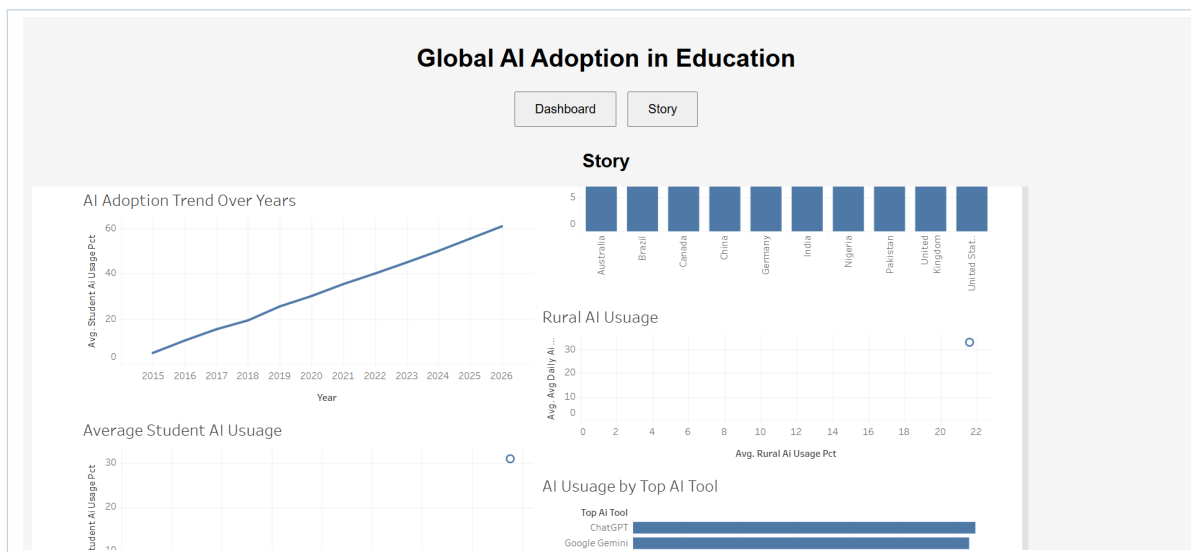


Figure 15: Flask web page with Dashboard and Story buttons

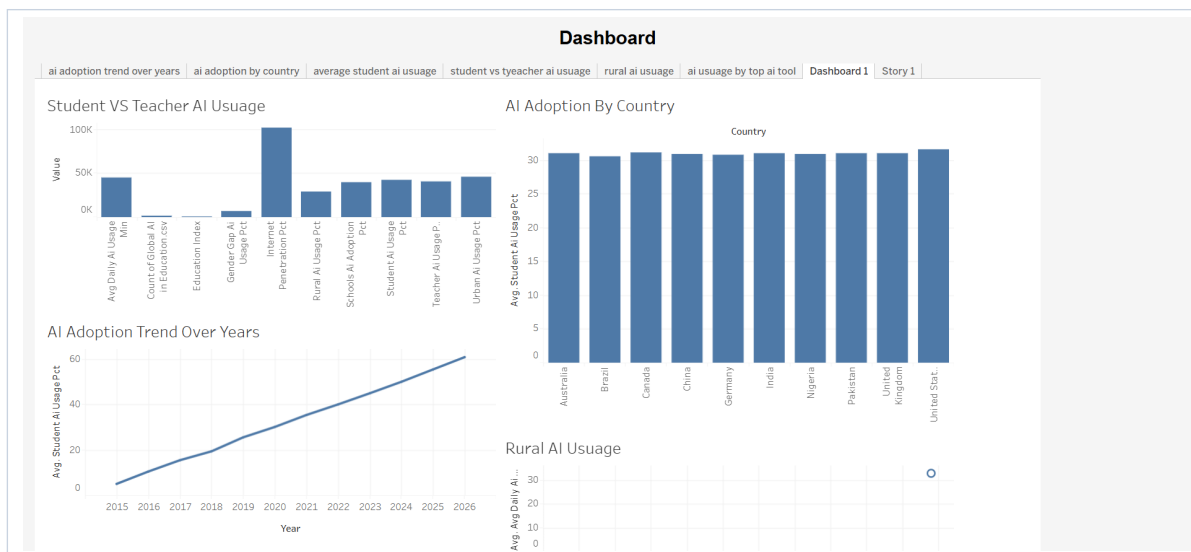


Figure 16: Tableau Dashboard displayed inside the Flask application

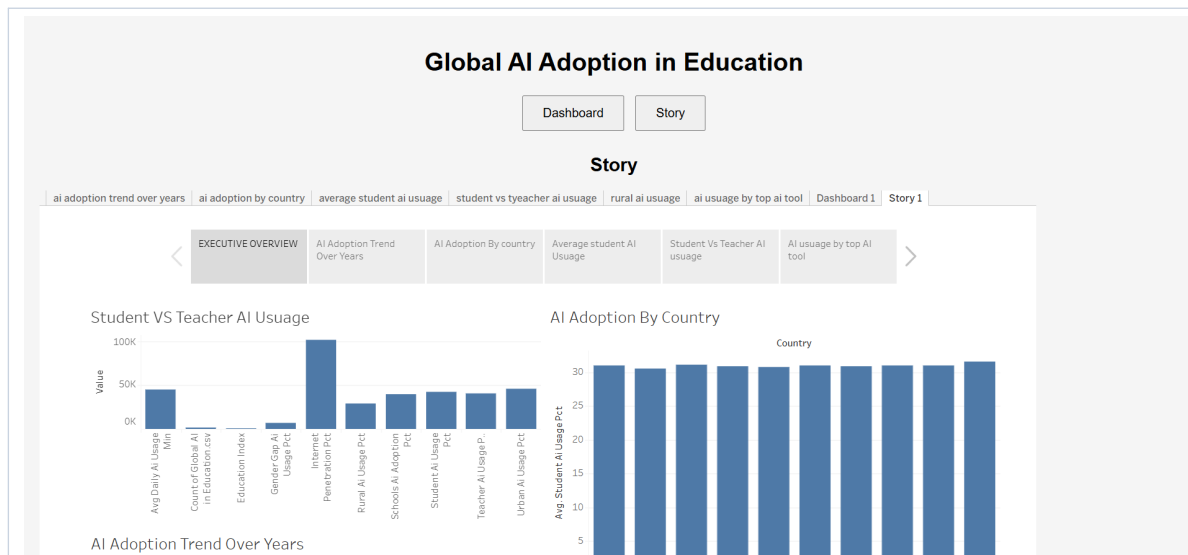


Figure 17: Tableau Story displayed inside the Flask application

6.2 Results / Key Insights

- Average student AI usage rose from about 5% in 2015 to about 61% in 2026 with no year of decline.
- All ten countries average about 30.6% - 31.6%; the United States is slightly highest.
- Students use AI slightly more than teachers (31.0% vs 29.9%).
- Urban usage (33.6%) is higher than rural usage (21.6%), so a rural gap exists.
- ChatGPT is the leading tool, followed closely by Google Gemini, Microsoft Copilot and Khanmigo.
- Average education index is about 0.78 and average AI use is about 33 minutes per day.

6.3 Limitations

The dataset is sample/synthetic, so findings demonstrate visualization and storytelling skills and are not verified global statistics. Averaging makes several charts show very small differences, and some scatter plots show only a single summary point. Figure 4 uses summed values of different measures, which are only meaningful as relative comparisons. The `government_ai_policy` field has missing values.

6.4 Conclusion

This project told the story of AI adoption in education through Tableau. Six visualizations were designed, combined into one dashboard and organised into a six-scene story, then published on Tableau Public and embedded in a Flask web application. The analysis shows steady growth in student AI usage, a small student lead over teachers, a clear rural-urban gap and ChatGPT as the most used tool. Although the data is synthetic, the project demonstrates a complete workflow from data to dashboard, story and web deployment.

6.5 Future Scope

- Replace the sample dataset with real survey and official education data.
- Add indicators such as impact on learning outcomes and barriers (cost, training, connectivity).
- Add state / city level maps and forecasting of future adoption.
- Connect Flask to a database for automatic updates; add login and report download.
- Deploy the Flask application on a cloud platform for public access.

6.6 References

- Tableau Public - public.tableau.com; Tableau documentation - help.tableau.com.
- Flask documentation - flask.palletsprojects.com.
- MDN Web Docs (HTML, CSS, JavaScript) - developer.mozilla.org.
- Dataset: Global AI in Education (CSV), sample / synthetic data.
- Published project: "Global ai Adoption in Education Final" by Sneha Yadav, Tableau Public.